6. Workflow Automation Using Deluge in Zoho Creator

Aim:

To automate Gym Membership enrollment and renewal activities using form workflows, Deluge scripting, schedules, batch processing, report actions and reusable functions, and review the available configuration and output evidence.

Definition:

A workflow executes actions when a specified event or condition occurs. Form workflows respond to data-entry events; schedules run at configured times; batch workflows process selected records; report actions respond to user commands; and custom functions centralize reusable logic. Deluge expresses validation, data access, email and other business actions.

Procedure:

1. Open the Gym Membership application and navigate to Workflow. Select Membership_Enrollment_Form.

2. Review initialization, enrollment-confirmation, enrollment-submit and status-timestamp workflows.

3. Configure age and preferred-start-date validation on form submission. The existing validation restricts age to 14-75 and rejects a start date in the past.

4. Configure the enrollment-submit workflow to check the email input before sending the welcome message. Review exception handling and the configured actions.

5. Submit a valid sample enrollment and inspect the received confirmation email. Test invalid inputs separately to verify rejected submissions.

6. Configure Daily_Renewal_Reminder_Schedule with its date/time, renewal-date criteria and email action. Verify the complete recurrence and reminder policy in its settings.

7. Create Bulk_Expiry_Notification_Batch with the required expiry criteria and monthly trigger. Review batch size and processing rules.

8. Create Generate_Membership_Card_Function with a member-record parameter, record-existence checks and required name, photo and email checks.

9. Expose a Send Membership Card report action and connect it to the intended member-record function call. Save the configuration and test with controlled sample records.

Automation Configuration and Evidence:

Component	Observed Configuration / Output
Form workflows	Enabled initialization, confirmation, enrollment-submit, status timestamp and validation workflows appear in the workflow list.
Enrollment submission	On Enrollment Submit Workflow has two actions. The visible script checks a non-null/non-empty email and begins a welcome sendmail action inside try handling.
Email output	A received Gym Membership Enrollment Confirmation email displays the plan, membership type, starting date, workout slot and joining instructions.
Schedules	Refresh Report Counts Daily and Daily Renewal Reminder Schedule are enabled. The latter shows 25-Oct-2026 at 09:00 as its displayed start.
Batch workflow	Bulk Expiry Notification Batch is enabled with a displayed trigger of 01-Nov-2026 at 09:00.
Custom function	The card function fetches by ID, rejects a missing record or required data, constructs HTML content and calls sendmail inside error handling.
Report actions	Open Members for Start Date and Send Membership Card Report Action are enabled as user-triggered report actions.

Testing and Debugging:

Check invalid age, a past starting date, empty email, unknown member ID and missing photo. Verify valid submissions and inspect logs for exceptions. Compare renewal selections at the date boundaries, confirm repeated runs do not exceed the intended reminder limit and test the report action with the correct record.

The received email demonstrates the existing confirmation action. Its subject differs from the newer welcome-script subject shown in the editor, so it should not be treated as proof of that newer action. Review the two enabled enrollment email workflows to avoid unintended duplicate messages.

Output - Form Workflows:

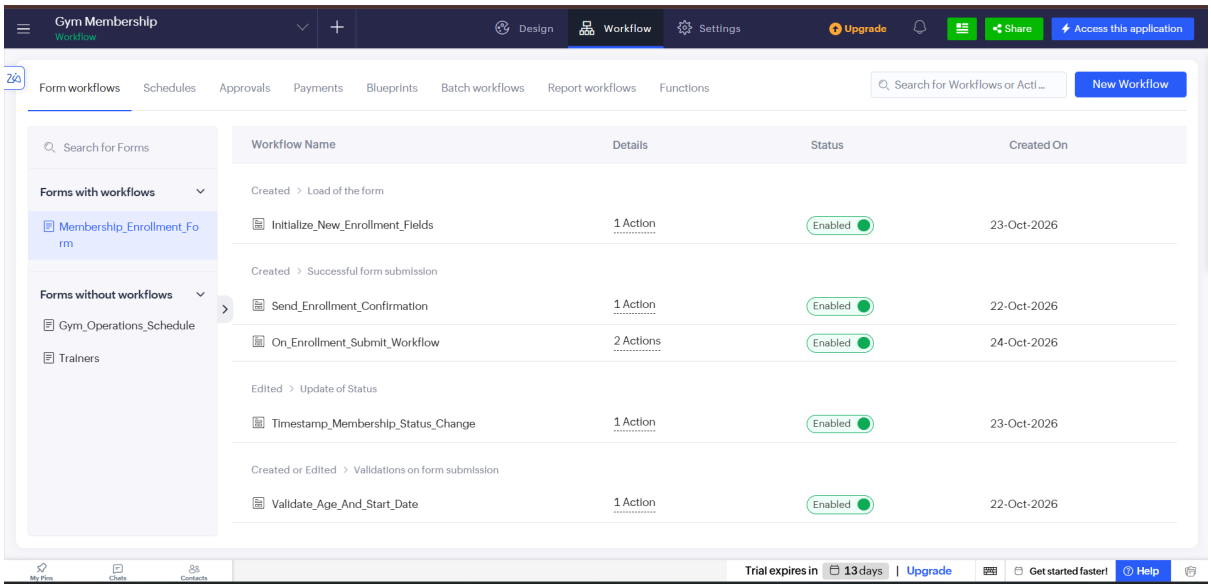


Figure 1. Enabled Gym Membership form workflows and their triggering events.

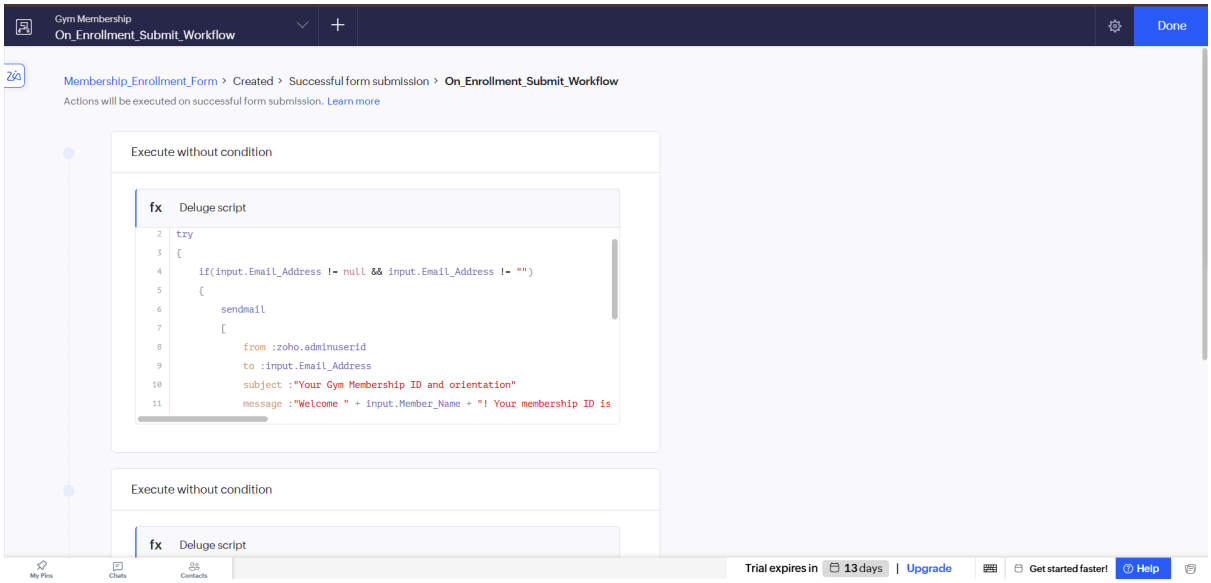


Figure 2. Enrollment-submit workflow showing email checks and welcome-message scripting.

Output - Confirmation and Renewal Schedule:

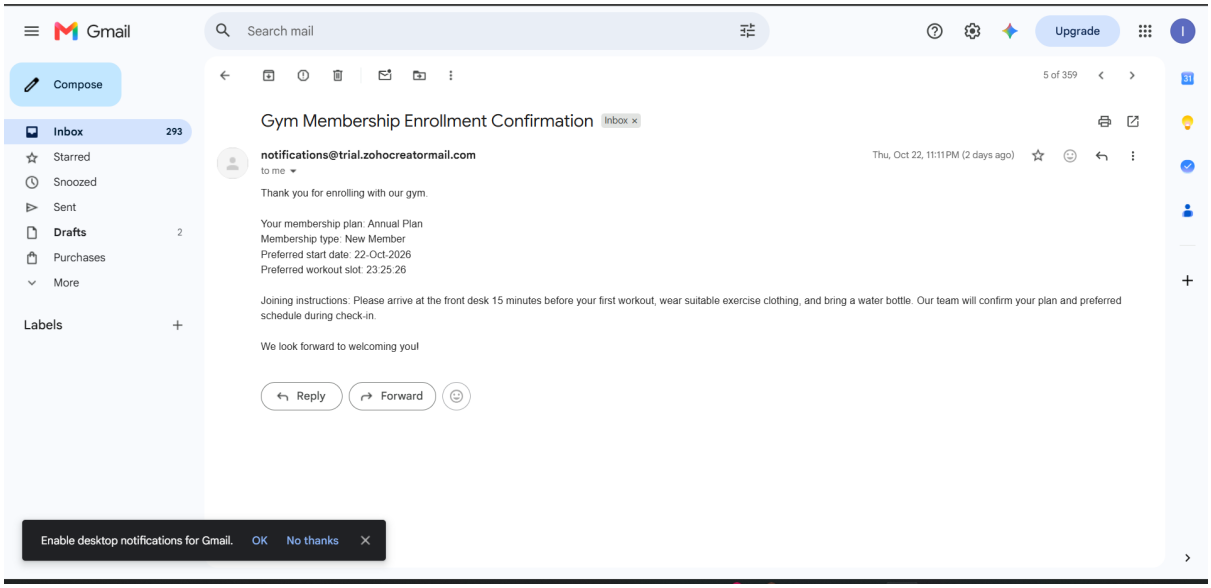


Figure 3. Received Gym Membership Enrollment Confirmation email with enrollment details.

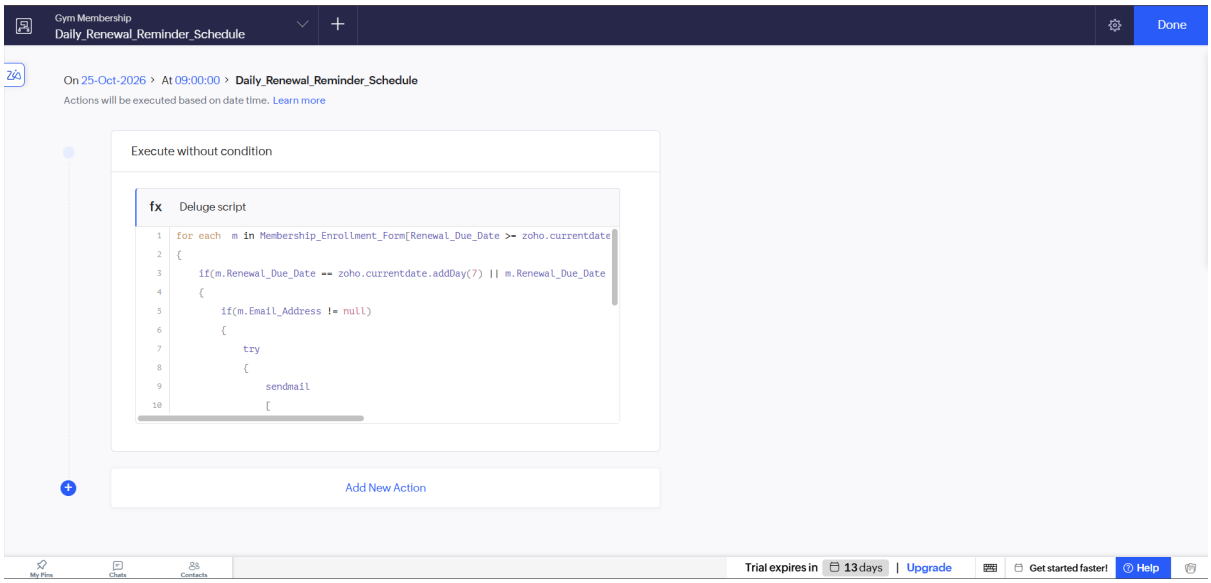


Figure 4. Daily Renewal Reminder Schedule editor and the visible renewal-date/email logic.

Output - Scheduled and Batch Automation:

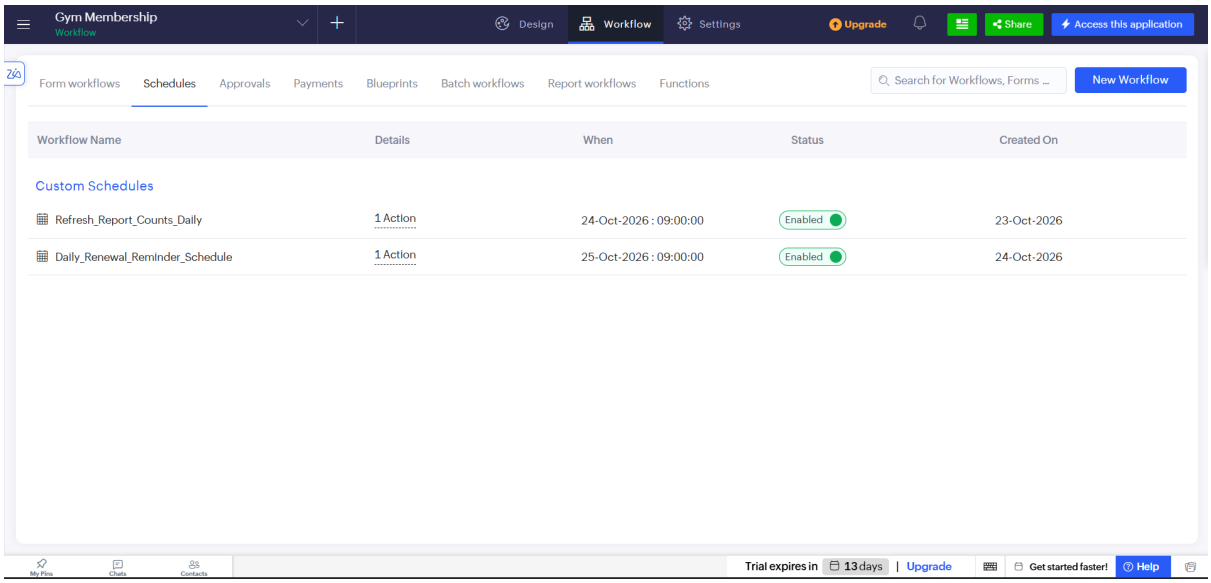


Figure 5. Enabled report-count and renewal-reminder schedules with displayed start times.

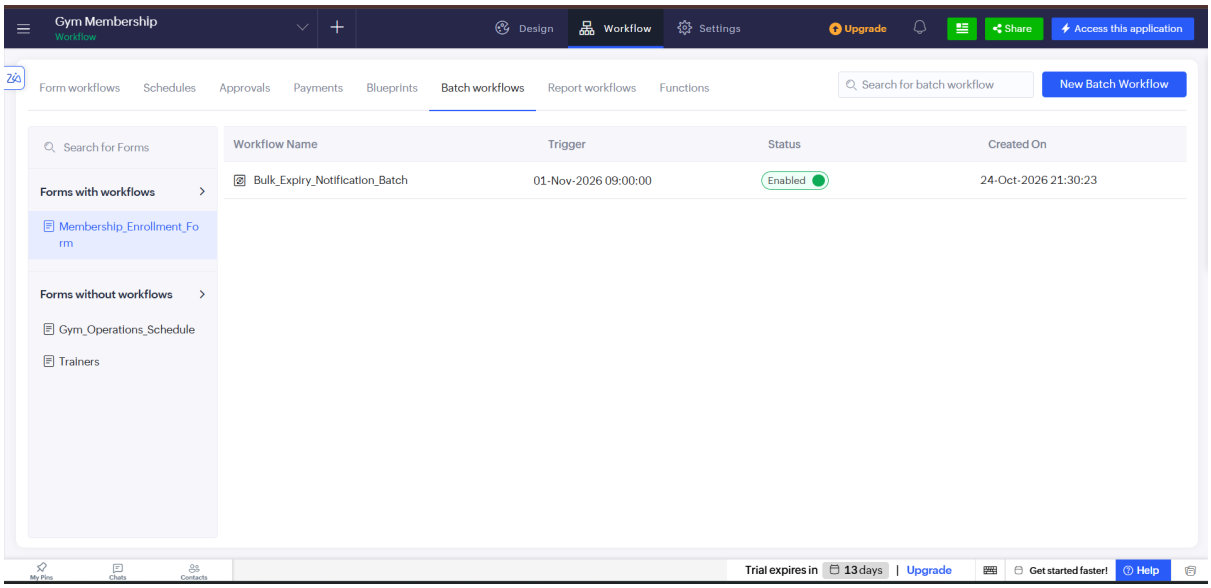


Figure 6. Enabled Bulk Expiry Notification Batch with its displayed trigger.

A configured future trigger establishes scheduling, not a completed run. The screenshots do not show the complete batch criteria, batch size, reminder cap or execution log.

Output - Custom Function and Report Actions:

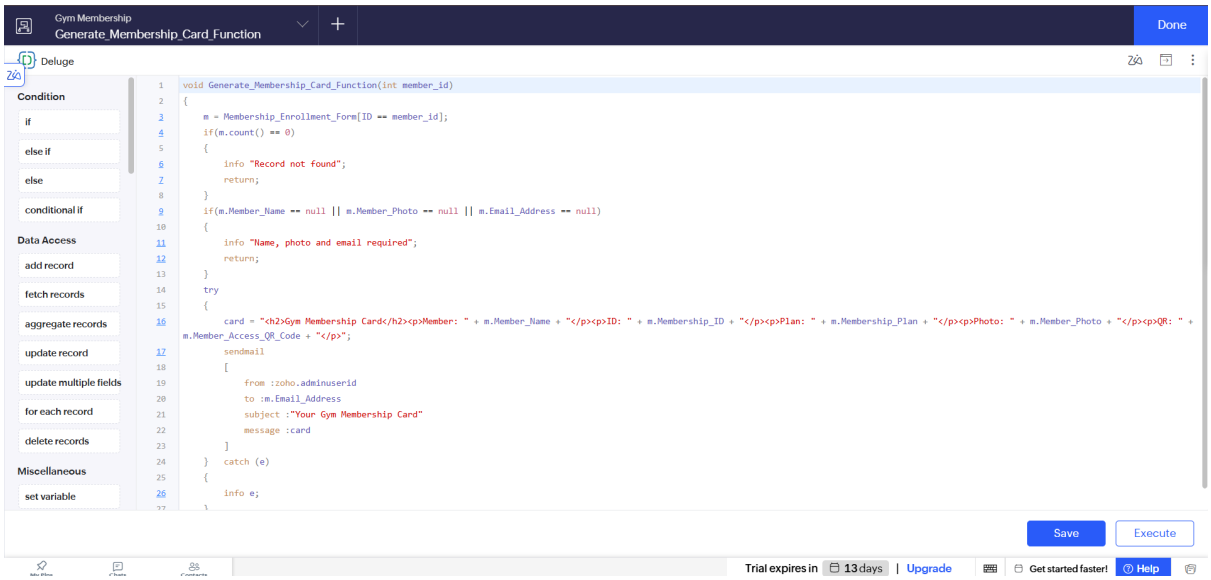


Figure 7. Generate Membership Card Function with record checks, HTML content and email action.

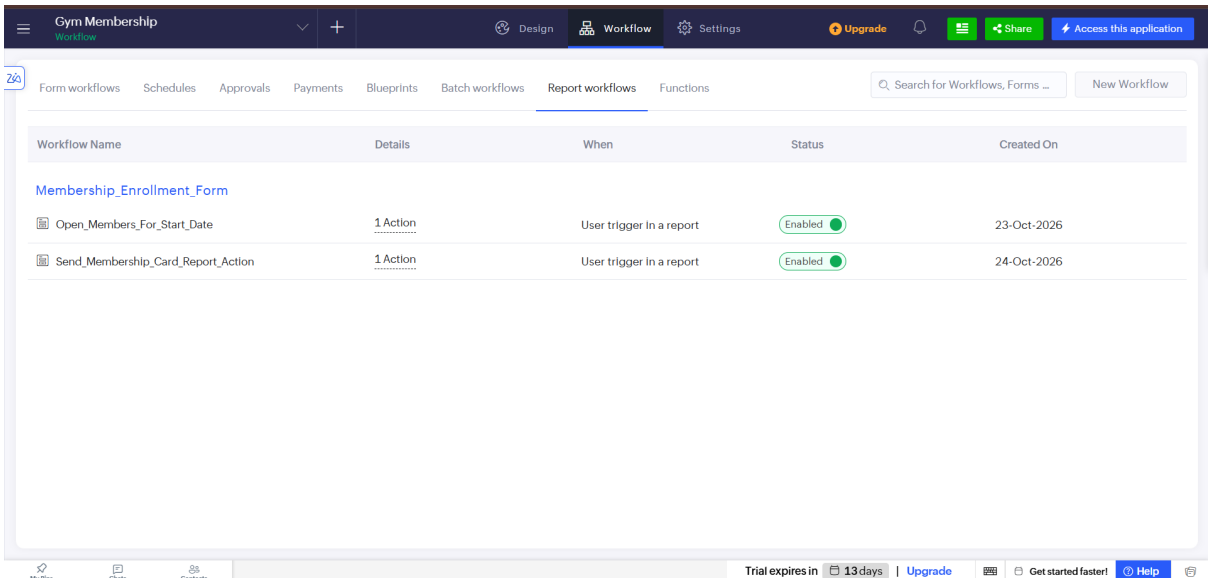


Figure 8. Enabled report actions for start-date navigation and sending a membership card.

Result:

Gym Membership automation was documented through form workflows, renewal scheduling, a batch definition, report actions and a Deluge membership-card function. A received email confirms an enrollment-notification output. Card generation/delivery, batch execution, approvals, payments and blueprint transitions are not demonstrated by the supplied images; the full eight-option assignment therefore requires additional verification.